Boolean Algebra

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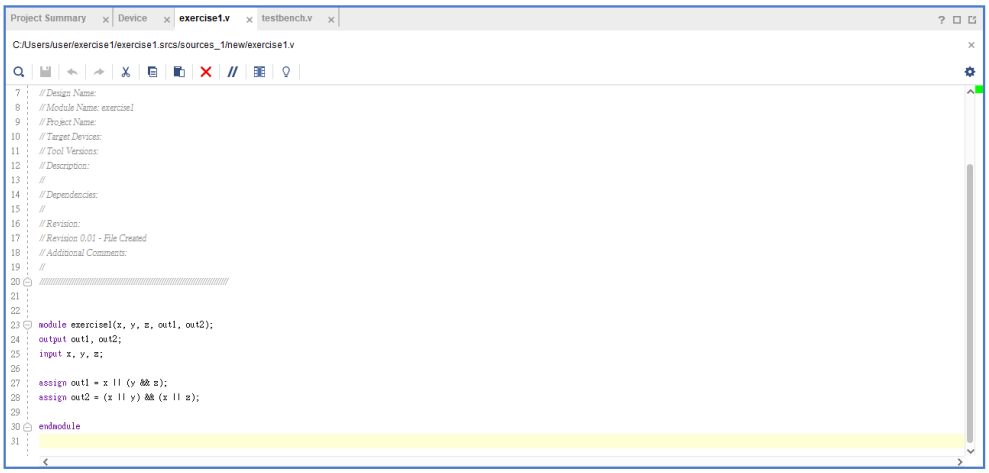
實驗內容

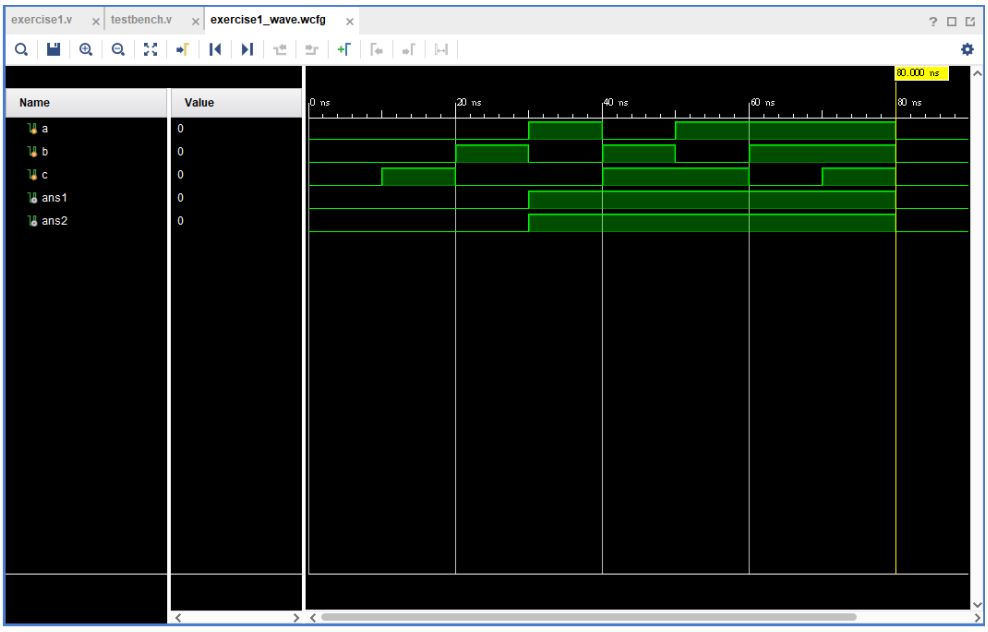
Exercise 1.

Verify Postulate 4(b): x + yz = (x + y)(x + z)

Let F1 = x + yz, and F2 = (x + y)(x + z).

Use **dataflow modeling** to verify that F1 = F2.





經由上述實驗結果可以發現，在使用布林代數的情況下，x + yz = (x + y)(x + z)。使用布林代數定理驗算過程如下：

x + yz = x(1 + z) + yz = x + xz + yz

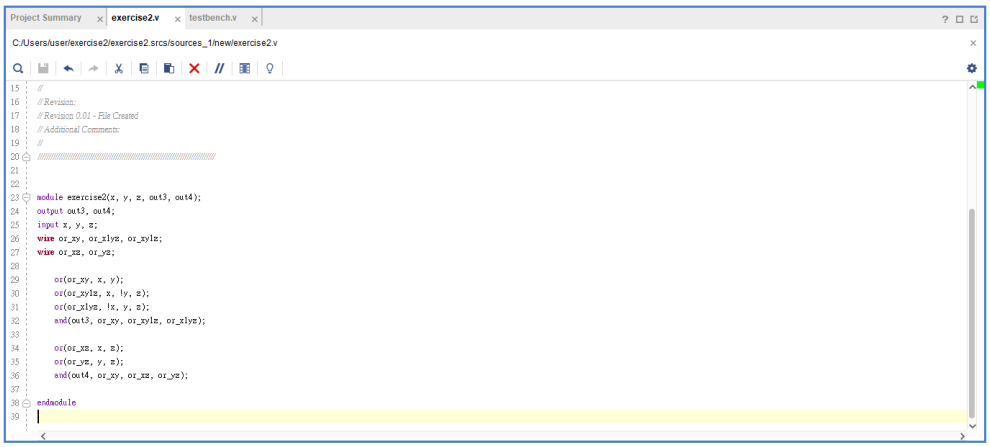
= x(1 + y) + xz + yz = x + xy + xz + yz

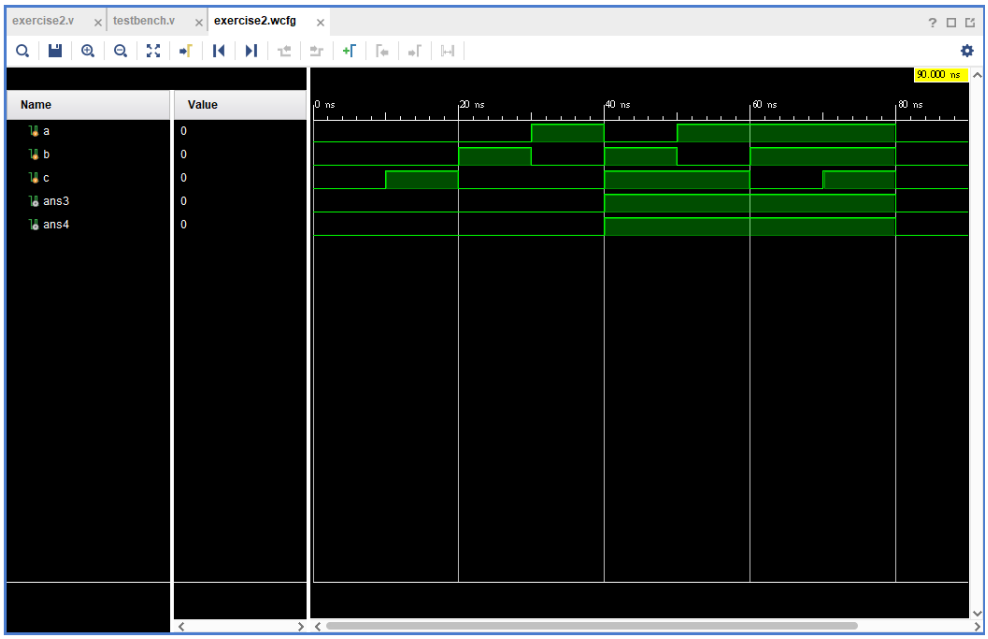
= xx + xy + xz + yz = **(x + y)(x + z)**

Exercise 2.

Let F3 = xy + xy'z + x'yz, and F4 = xy + xz + yz.

Use **structural level modeling** to verify that F3 = F4.





經由上述實驗結果可以發現，在使用布林代數的情況下，xy + xy'z + x'yz = xy + xz + yz。使用布林代數定理驗算過程如下：

xy + xy'z + x'yz = xy(1 + z) + xy`z + x`yz

= xy + xz(y + y`) + x`yz

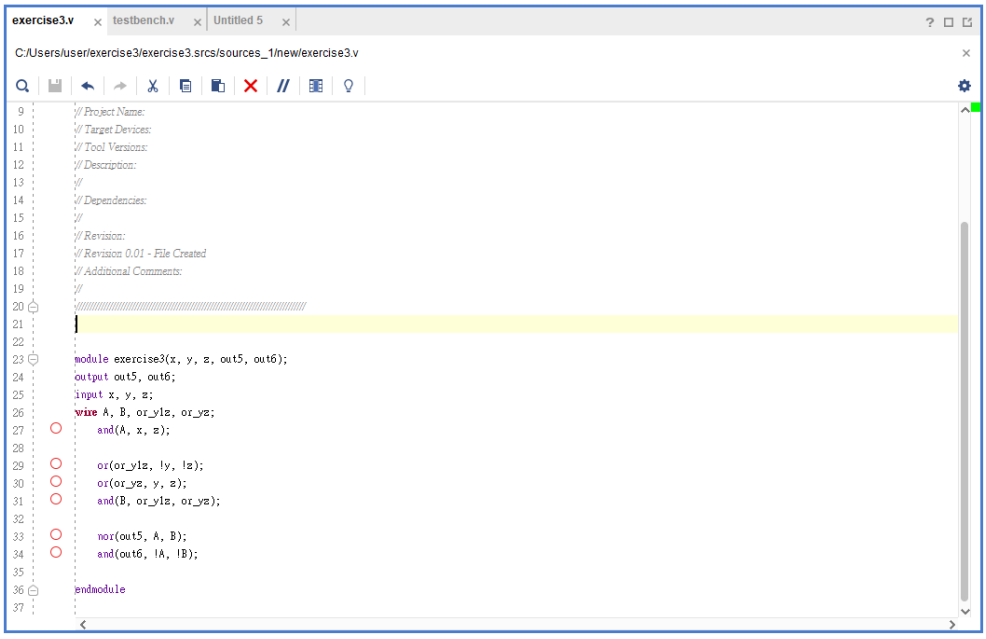
= xy + xz + x`yz = xy + xz(1 + y)+ x`yz

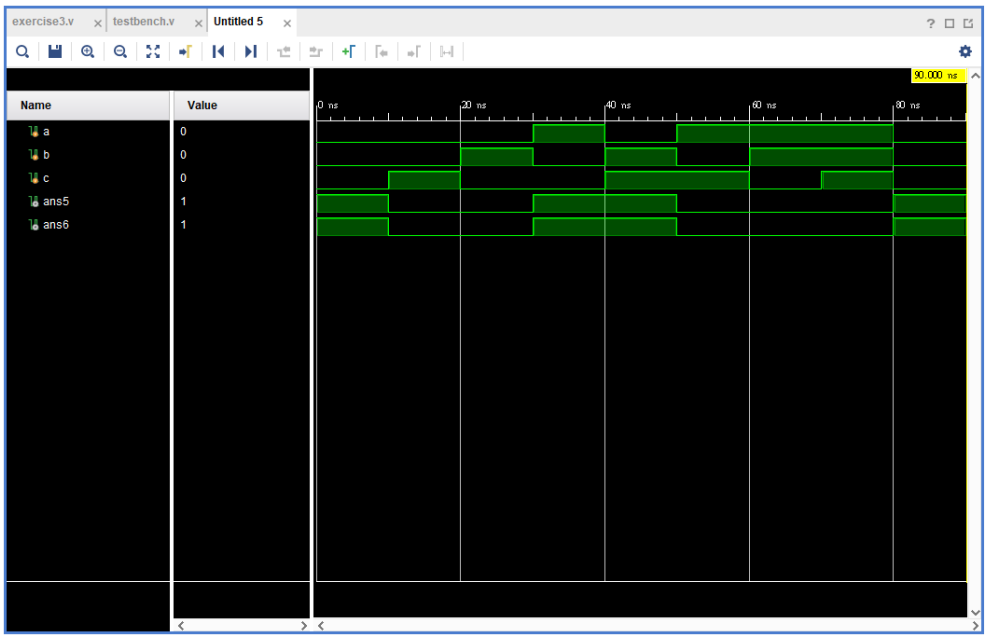
= xy + xz + xyz + x`yz = **xy + xz + yz**

Exercise 3.

Let F5 = (x + z)(y'z' + yz).

Find and show the complement of F5 (i.e., F6) by using **DeMorgan's theorem**, and use **structural level modeling** to verify that F5' = F6.





Let F5 = (x + z)(y'z' + yz) = AB, where A = x+z, B = y'z' + yz.

$\Longrightarrow$ F5' = (AB)' = [(x + z)(y'z' + yz)]'

By DeMorgan's theorem, (AB)' = A' + B'.

$\Longrightarrow$ F6 = A' + B' = (x + z)' + (y'z' + yz)'

同時，經由上述實驗結果也可以驗證，在使用布林代數的情況下，F5' = [(x + z)(y'z' + yz)]' = (x + z)' + (y'z' + yz)' = F6。

實驗心得

總體來說，這次實驗進行得比上次好很多，可能是因為已經對 vivado 和 verilog 有初步熟悉的關係。

這次實驗的主要內容是以撰寫 verilog 來驗證在上課時所提到的布林代數定理，以及 Demorgan's theorem。我覺得當中最困難的部分在於陌生的定理這

部分，因為突然冒出太多定理了，大腦沒法在短時間內記起來，在撰寫程式時還要時常去翻上課的 ppt，不然根本清楚這個定理得推倒過程或是這個定理在說甚麼。看來之後要多加複習，至少要把 Demorgan's theorem 給背熟。